

Sri Sivasubramaniya Nadar College of Engineering, Kalavakkam
(An Autonomous Institution, Affiliated to Anna University, Chennai)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Date: 07-08-2023

Course Code	UCS2511	Course Name	Networks Lab					
Course Type	Lab	Course Category	Professional Core (PC)	L	T	P	E	C
				0	0	3	0	1.5
Regulation	2021		Academic Year & Batch	2024 - 25 & 2022 - 26				
Degree and Branch			B.E. Computer Science & Engineering					
Semester			V					
Department Offering the Course			Computer Science and Engineering					
Name of the Faculty			Dr. J. Bhuvana & Dr. N. Sujaudeen					

OBJECTIVES

- To learn and use network commands.
- To learn socket programming.
- To implement and analyze various network protocols.
- To learn and use simulation tools.
- To use simulation tools to analyze the performance of various network protocols.

SUGGESTIVE EXPERIMENTS

1. Learn to use commands like tcpdump, netstat, ifconfig, nslookup and traceroute. Capture ping and traceroute PDUs using a network protocol analyzer and examine.
2. Write applications using TCP sockets like:
 - Echo client and echo server
 - Chat
 - File Transfer
3. Simulation of DNS using UDP sockets.
4. Write Programs to implement flow control protocol in data link layer.
5. Simulation of error detection code (like CRC).
6. Write programs simulating ARP /RARP protocols.
7. Study of Network simulator (NS) and simulate a wired topology using NS.
8. Simulate Distance Vector/ Link State Routing algorithm.
9. Study of TCP/UDP performance using Simulation tool.
10. Simulation of Congestion Control Algorithms using NS2 and analyse its performance.

TOTAL PERIODS (LAB): 45

Course Outcomes:

1. Develop socket programs using TCP and UDP, to implement various application protocols (K3)
2. Develop socket programs to implement the functionalities of various layers in TCP / IP protocol suite (K3)
3. Analyze the performance of various protocols using a network simulator tool. (K4)

LAB PLAN

S.No	List Of Experiments	K Level	CO	PI	No of Practical Sessions		Deviation
					Proposed hours	Actual hours	
1	Study of commands and system calls for simple client server program	K2	CO1	1.3.1, 1.4.1	1		
2	Simple Client Server connection, Echo client Server	K3	CO1	1.4.1	1		
3	File transfer using TCP	K3	CO1	2.1.2, 4.1.3	1		
4	Chat Application	K3	CO1	2.3.1, 4.1.2, 13.3.1	1		
5	Simulation of ARP / RARP protocol using TCP	K3	CO2	1.3.1, 1.4.1, 4.1.2	1		
6	Simulation of DNS protocol using UDP	K3	CO2	2.1.2, 4.3.1	2		
7	Simulation of flow control protocol in data link layer	K3	CO2	1.4.1, 4.3.1, 13.3.1	1		
8	Simulation of error correction mechanism	K3	CO2	1.4.1, 4.3.1, 13.3.1	2		
9	Study of Network simulator NS	K2	CO3	1.3.1, 5.2.2	1		
10	Simulation of Routing protocol in NS2	K3	CO3	2.1.2, 4.1.2, 5.2.2	2		
11	Performance analysis of TCP and UDP connections sharing a bottleneck link	K4	CO3	1.4.1, 4.3.1, 5.2.2, 13.3.1	1		
12	Performance analysis of Congestion Control Algorithms	K4	CO3	1.4.1, 4.3.1, 5.2.2, 13.3.1	1		

Total Sessions: 45 sessions

CO-PO/PSO mapping using Performance Indicators

PO/ PSO	CO1		CO2		CO3	
	PI	PI-MS	PI	PI-MS	PI	PI-MS

1	1.3.1 (3) 1.4.1 (3)	3	1.3.1 (3) 1.4.1 (3)	3	1.3.1 (3) 1.4.1 (3)	3
2	2.1.2 (3) 2.3.1 (3)	3	2.1.2 (3) 2.3.1 (3)	3	2.1.2 (3) 2.3.1 (3)	3
3	-	-	-	-	-	-
4	4.1.2 4.3.1	3	4.1.2 4.3.1	3	4.1.2 4.3.1	3
5	-	-	-	-	5.2.2 (2)	2
6	-	-	-	-	-	-
7	-	-	-	-	-	-
8	-	-	-	-	-	-
9	-	-	-	-	-	-
10	-	-	-	-	-	-
11	-	-	-	-	-	-
12	-	-	-	-	-	-
13	13.3.1 (2)	2	13.3.1 (2)	2	13.3.1 (2)	2
14	-	-	-	-	-	-

CO to PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	-	3	-	-	-	-		-	-	-	2	-
CO2	3	3	-	3	-	-	-	-		-	-	-	2	-
CO3	3	3	-	3	2	-	-	-		-	-	-	2	-
Total	9	9	-	9	2	-	-	-		-	-	-	6	-
Score	3	3	-	3	2	-	-	-		-	-	-	2	-

PI	Description	PI strength
1.3.1	Apply engineering fundamentals	3
1.4.1	Apply theory and principles of computer science and engineering to solve engineering problem	3

2.1.2	Identify processes/modules/algorithms of computer-based systems and parameters to solve problems	3
2.3.1	Apply computer engineering principles to build modules of a system with required applicability and performance	3
4.1.2	Choose appropriate procedures/algorithms, datasets, and test cases.	3
4.3.1	Use appropriate procedures, tools, and techniques to collect and analyze data	3
5.2.2	Demonstrate proficiency in using computer science and engineering tools	2
13.3.1	Apply fundamentals of problem solving	3

Assessment Tools & Evaluation Criteria:

Assessment Tool	Assessment Technique	Weightage	Marks
Lab Exercises	Best Practices	15	35%
	Algorithm	20	
	Implementation	45	
	Output	20	
Lab Test	Best Practices	15	10%
	Algorithm	20	
	Implementation	45	
	Output	20	
Model Exam	Best Practices	5	15%
	Algorithm	15	
	Implementation	45	
	Output	25	
	Viva	10	
End Semester Exam	Best Practices	5	40%
	Algorithm	15	
	Implementation	45	
	Output	25	
	Viva	10	

Prepared By	Verified By	Approved By
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